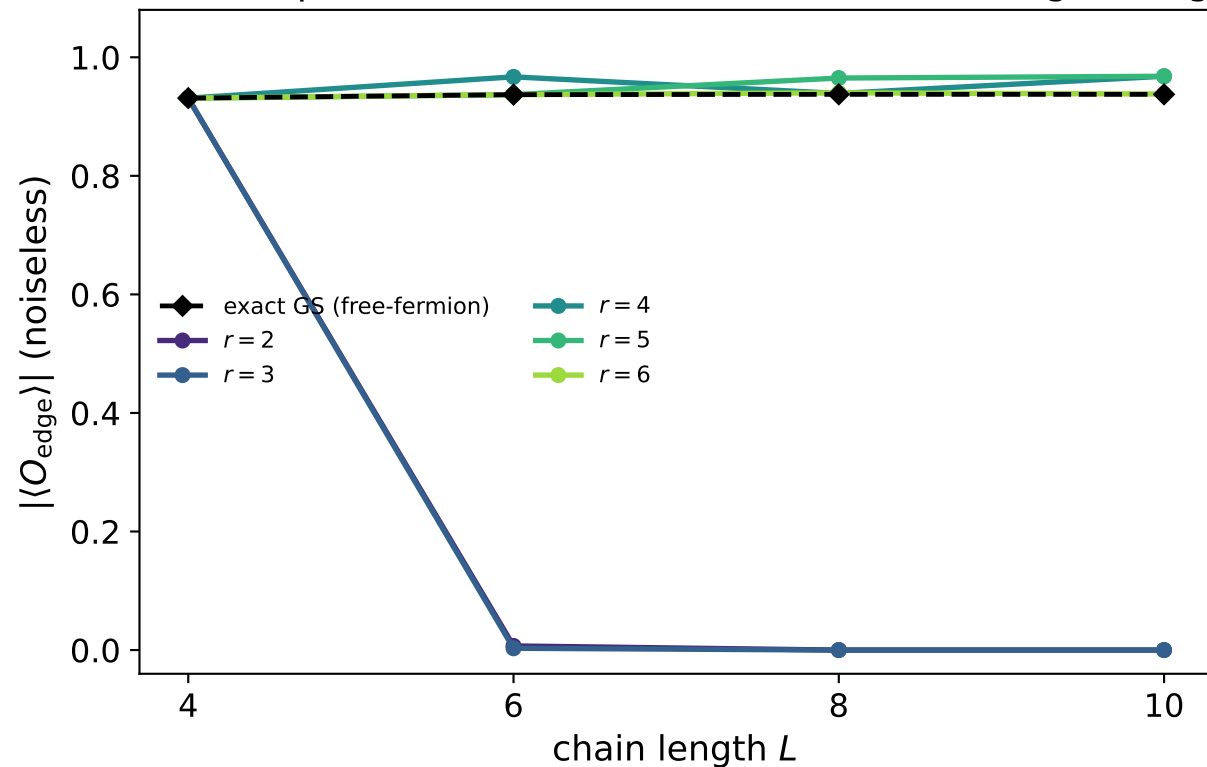


Two ways a fixed-depth preparation fails as L grows ($\mu = 0.5t$, $p_{\text{CX}} = 0.05$)

Fixed depth: shallow circuits cannot build the edge string



Minimal adequate depth: noise then wins

